

NGINX Ingress Controller — Internal & External

Nexlify Technologies | Controllers: nginx-ext, nginx-int | ap-south-1

1. Dual-Controller Architecture

Nexlify deploys two separate NGINX Ingress Controller instances inside each EKS cluster. The external controller (release name `nginx-ext`) is exposed via an internet-facing AWS Application Load Balancer and terminates public HTTPS traffic for customer-facing hostnames. The internal controller (release name `nginx-int`) is exposed via an internal ALB and serves private hostnames resolvable only inside the VPC or via the corporate VPN. Both controllers run as Deployments with HPA (min 2, max 8 replicas in production). They use distinct IngressClass resources (`nginx-external` and `nginx-internal`) so that application Ingress objects explicitly select the desired entry point. Mixing classes on a single hostname is forbidden.

2. External Ingress Controller Details

The external controller listens on ports 80 and 443. TLS certificates are provisioned by `cert-manager` using Let's Encrypt (HTTP-01 for non-prod, DNS-01 for production). Default SSL policy is AWS-TLS-1-2-2021-06 or stronger. Rate limiting at the ingress layer is kept conservative; sophisticated rate limiting is delegated to Kong (see Kong runbook). WAF association is performed on the ALB via AWS WAFv2 web ACL that includes AWS Managed Rules and a custom geo-blocking rule set. Access logs are shipped to S3 and indexed in OpenSearch. Annotations for external Ingress objects must include the correct ingress class and may set `proxy-body-size`, `proxy-read-timeout`, and canary settings when needed.

3. Internal Ingress Controller Details

The internal controller is the preferred entry point for service-to-service traffic that originates inside the VPC (e.g., batch jobs, internal admin UIs, inter-team APIs). It does not have a public IP. Certificates are issued by the internal ACM Private CA via `cert-manager`. Authentication is typically handled by the service mesh or by Kong; the internal NGINX layer focuses on routing, path rewriting, and basic header manipulation. Because internal traffic never leaves the VPC, data-transfer costs are lower and exposure surface is reduced. Teams are encouraged to use internal hostnames (`*.internal.nexlify.io`) for any non-public endpoint.

4. Operational Procedures

Controller upgrades are performed via Helm (or Argo CD) after testing in non-prod. Configuration changes to ConfigMaps (`nginx.conf` snippets, lua plugins) require a PR to the platform-gitops repository. Canary releases of new ingress configurations use the `nginx.ingress.kubernetes.io/canary` annotations. When an ALB target group becomes unhealthy, the AWS Load Balancer Controller automatically deregisters the target; the NGINX pods themselves are protected by a PDB (`maxUnavailable: 1`). Metrics are scraped by Prometheus from the `/metrics` endpoint; key alerts include high 5xx rate, upstream latency, and configuration reload failures.

5. Common Annotations Reference

• `kubernetes.io/ingress.class: nginx-external | nginx-internal` • `nginx.ingress.kubernetes.io/ssl-redirect: true` • `nginx.ingress.kubernetes.io/proxy-body-size: 10m` • `nginx.ingress.kubernetes.io/proxy-read-timeout: 60` • `nginx.ingress.kubernetes.io/canary: true` (with `canary-weight`) • `external-dns.alpha.kubernetes.io/hostname` for automatic Route 53 records.

Owner: Traffic Platform | Slack: `#ingress-ops` | Rev 3.0 | 2026-05-30